

COMPSCI 4CR3: Applied Cryptography

Department of Computing and Software
McMaster University

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Lectures:	Wednesday 11:30 - 12:20, Friday 10:30 - 12:20 BSB B135 (Wednesday), HH 302 (Friday)
Office Hours:	Thursday (By appointment)

Course description

The ubiquity of communication devices and the increasing need for data protection have made information security an indispensable part of our daily lives. Cryptography, which originated long before the advent of computers, has evolved significantly from its early forms of encryption. Today, modern cryptography encompasses a wide array of techniques far beyond simple encryption. This course serves as an introduction to cryptography, focusing on its practical applications. We will explore three core topics:

- Symmetric Cryptography: We will discuss stream and block ciphers, examining how they provide confidentiality in various contexts.
- Hash Functions: This section will cover foundational concepts, highlighting their critical role in data integrity, with a detailed look at the SHA-2 algorithm as a real-world example.
- Asymmetric Cryptography: We will discuss essential definitions, explore the RSA encryption algorithm, and understand how digital signatures ensure authenticity and non-repudiation in communication.

- *Prerequisites:* COMPSCI 2C03 or SFWRENG 2C03.

- *Textbooks:* The lectures will be mainly based on the following book:
 - “Understanding Cryptography” by Christof Paar and Jan Pelzl.

- *Organization:* 3 hours of lecture, 1 tutorial.

- *Learning Objectives:* At the end of this course, students will
 - have a reasonable understanding of some of the basic ideas in information security
 - understand the basic mathematical and practical aspects of cryptographic algorithms,
 - be able to explain basic attacks on cryptographic protocols.

- *Tentative list of topics:*
 - An Introduction to Cryptography
 - Stream Ciphers
 - The Data Encryption Standard (DES)

- The Advanced Encryption Standard (AES)
- Hash Functions
- Introduction to Public-Key Cryptography
- The RSA Cryptosystem
- Digital Signatures

Evaluation

The final grade for this course will be calculated based on the grades from assignments, quizzes, and the final exam. There will be four assignments and a final exam. The first two assignments will focus on topics related to block ciphers and hash functions, while assignments 3 and 4 will cover public-key cryptography. The final exam will assess knowledge of the entire course.

Evaluation	% of the final grade	Topic
Quizzes	5%	During class
Assignment 1	15%	Block ciphers and hash functions
Assignment 2	15%	
Assignment 3	15%	Public-key cryptography
Assignment 4	15%	
Final project	35%	Entire course

Evaluation details:

- Quizzes will be given without notice. Quizzes should be submitted by the end of the same day.
- The assignments will be spread out evenly throughout the semester.
- Late submissions will be penalized at a rate of $2^n\%$, where n is the number of days late. One day late is a 2% penalty, two days 4%, three days 8%, and so on.
- We will use **Avenue to Learn** for course management. All course materials can be accessed through Avenue to Learn. Other things such as announcements, assignment descriptions and submissions, etc, will also be done through Avenue to Learn.
- Dates are subject to change, all changes will be announced in Avenue to Learn.

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: “Grade of F assigned for academic dishonesty”), and/or suspension or expulsion from the university.

For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/), located at <https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/>.

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity/Plagiarism Detection

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Courses With An On-Line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in

virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

Academic Accommodation Of Students With Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's [Academic Accommodation of Students with Disabilities](#) policy.

Requests For Relief For Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).¹

Academic Accommodation For Religious, Indigenous Or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright And Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.

¹Updated to the Policy on Requests for Relief for Missed Academic Term Work effective May 1, 2021